PRAHARI NETRA

Professional Border Surveillance Web App — Master Build Brief for Claude

VISUAL IDENTITY	**Military green + beige** · disciplined · premium · secure · Indian security-operations aesthetic

1. CORE PRODUCT VISION

Build **Prahari Netra** as a professional border-surveillance and evidence-intelligence platform, not a generic dashboard. Its differentiator is the combination of AI video analytics, a virtual India map for camera exploration, role-specific workspaces, and cryptographic evidence integrity.

- **AI surveillance:** detect and classify relevant objects/events from existing CCTV/RTSP feeds.
- **Geospatial command:** an interactive India map lets an authorized admin virtually move to a border sector and inspect cameras.
- **Evidence integrity:** show SHA-256 verification, digital-signature status, certificate/trust status and tamper-evident event history.
- **Role separation:** Admin Panel and Ground Officer Panel must look and behave differently.
- **Demo-first:** realistic simulated cameras/events so the UI works without physical CCTV hardware.

2. FIRST SCREEN — INDIA MAP + PRAHARI NETRA

When the app opens, do **not** immediately show a conventional dashboard. The landing screen should be a full-screen India map with a premium military-operations aesthetic.

- Full-screen stylized India map as the hero/background.
- Brand name **PRAHARI NETRA** with subtitle *Border Intelligence & Surveillance Network*.
- Subtle border-sector markers, camera nodes and status indicators; no clutter.
- Muted military-green / olive map treatment on beige/cream terrain.
- Top-left logo/wordmark; top-right system status such as SECURE NETWORK / SYSTEM OPERATIONAL.
- Strong **LOGIN** button with restrained animation.
- Subtle node pulses/grid effects; absolutely no hacker/matrix styling.
- Clearly label prototype/demo capabilities; do not imply live classified government access.

3. LOGIN → TWO DISTINCT PANELS

Click LOGIN → clean authentication screen → after demo authentication show two role choices. The panels must be visually and functionally different.

ADMIN PANEL	GROUND OFFICER PANEL
Command, geography, cameras, analytics, users, system health, evidence audit.	Assigned sector, live alerts, camera feeds, incident response and evidence capture.

Implement role-based access control. Users should only see actions permitted by their role.

4. ADMIN PANEL — COMMAND CENTER

- **Overview:** cameras online/offline, active incidents, unresolved alerts, evidence-integrity health and AI events.
- **India Operations Map:** signature feature. Admin pans/zooms, selects a sector/location, then opens nearby cameras.
- **Camera Explorer:** camera ID, simulated location, live/simulated stream, health, last heartbeat and AI detections.
- **Incidents:** searchable timeline with severity, sector, camera, timestamp, event type, confidence and evidence status.

- **Evidence Vault:** SHA-256, signature, certificate and hash-chain verification details.
- **Analytics:** event trends, camera uptime, alert review and sector activity.
- **System Health:** connectivity, AI latency, API health, storage and processing status.
- **Access Control:** users, roles, permissions and audit logs.

5. INDIA MAP — THE DIFFERENTIATOR

Create a dedicated **Geospatial Command Map**. This should be one of the strongest screens in the prototype.

- Center the map on India with visually distinct border sectors.
- Camera markers: ONLINE, OFFLINE, ALERT, MAINTENANCE.
- Click state/sector → zoom in → show nearby camera nodes.
- Click camera → side panel with Camera ID, sector, demo coordinates, status, heartbeat and AI events.
- **Open Camera** action opens camera detail/live feed.
- Filters: All Cameras, Online, Alert, Offline, AI Activity, Evidence Compromised.
- Use fictional/demo locations and coordinates; never claim live government data.
- Support local/static India-map fallback so the demo works without internet.

6. GROUND OFFICER PANEL — DIFFERENT UX

Do not duplicate the Admin dashboard. Ground Officer is faster, simpler and field-oriented.

- **My Sector:** assigned border sector and nearby cameras.
- **Live Watch:** camera grid with AI boxes, class, confidence, timestamp and health.
- **Alert Queue:** priority events with acknowledge / investigate / escalate.
- **Incident Detail:** timeline, evidence, location, detection metadata and integrity status.
- **Evidence Capture:** create a demo evidence package and verify its cryptographic status.
- **Map:** operational geography relevant to the assignment.
- No user-management or heavy administrative controls.

7. CAMERA DETAIL PAGE

- Large live/simulated CCTV feed.
- AI overlay: object boxes, class, confidence and timestamp.
- Metadata: Camera ID, sector, location, status, FPS, latency and last heartbeat.
- AI event timeline.
- Evidence badge: **VERIFIED / WARNING / COMPROMISED**.
- Actions: Investigate Event, View Evidence, Verify Integrity, Open Map Location.
- Playback/demo mode must work without real CCTV.

8. EVIDENCE INTEGRITY — TRUST LAYER

Make the cybersecurity layer visible and technically meaningful. Do not present hashing as encryption.

- **SHA-256:** fingerprint of evidence.
- **Digital signature:** proves the evidence/hash was signed by a trusted source.
- **Certificate / PKI:** establishes trust in camera/edge identity.
- **mTLS:** authenticates and protects machine-to-machine communication.
- **Hash chain:** makes sequential evidence/event history tamper-evident.
- **JWT + RBAC:** human authentication/authorization; JWT is not the video-integrity mechanism.

Demo: verified evidence shows **INTEGRITY VERIFIED**. A controlled tamper action changes the file; re-hashing then displays **HASH MISMATCH** → **SIGNATURE INVALID** → **EVIDENCE COMPROMISED**. Clearly label this as a controlled demonstration.

9. VISUAL DESIGN SYSTEM

Element	Direction
Primary	Deep military / forest green
Secondary	Olive / muted sage
Background	Warm beige / parchment / cream
Cards	Light beige with subtle green borders
Text	Deep charcoal / dark green
Critical	Restrained red only for critical alerts
Warning	Muted amber
Verified	Operational green

- Modern readable sans-serif; uppercase only for operational labels.
- Restrained corners and shadows; avoid generic rounded SaaS-card styling.
- Consistent line icons.
- Subtle transitions/map pulses only.
- Overall feeling: serious security-operations product, not a game or hacker dashboard.

10. TECHNICAL TARGET

- **Frontend:** React + Vite + TypeScript.
- **Styling:** Tailwind CSS or clean component system with centralized green/beige tokens.
- **Map:** Leaflet or MapLibre/Mapbox-compatible architecture with local/static fallback.
- **Backend:** FastAPI/Python.

- **Auth:** JWT + RBAC.
- **AI:** modular inference service with prerecorded/simulated CCTV fallback.
- **Streaming:** RTSP/WebRTC-ready abstraction plus local playback.
- **Database:** PostgreSQL recommended; SQLite acceptable for initial local prototype.
- **Integrity:** SHA-256 + signatures + certificate metadata + hash-chain records.
- **Architecture:** separate UI, API, AI inference, evidence/integrity and data layers.

11. DEMO DATA / RELIABILITY

Seed 8–15 fictional camera nodes across Indian border sectors, normal events, alerts and at least one compromised-evidence example. Add a visible **DEMO MODE** indicator. The entire presentation must work without physical CCTV hardware.

12. CLAUDE MASTER PROMPT

Paste this PDF together with the following instruction:

```
You are a senior product designer and full-stack engineer. Build the Prahari Netra application exactly from this
specification. Do not merely describe it: create the working project. Start with the complete frontend shell and
navigation, then implement the India-map command experience, role-specific Admin and Ground Officer panels, camera
detail pages, incidents, evidence-integrity UI, demo data and responsive behavior. Use military green and beige as
the dominant palette. Make the product premium, minimal, operational and trustworthy. Avoid hacker/matrix aesthetics.
Make every navigation path and button functional. Use fictional/demo data where real infrastructure is unavailable.
Explain major technology choices and give exact terminal commands to install, run and test. Build in small verifiable
stages and keep the code production-structured.
```

13. HACKATHON DEMO FLOW

- Open → India map hero → PRAHARI NETRA.
- LOGIN → choose Admin Panel.
- Geospatial Command Map → border sector → camera.
- Open Camera → AI detection → incident alert.
- Evidence → cryptographic verification.
- Controlled tamper demo → HASH MISMATCH / EVIDENCE COMPROMISED.
- Switch to Ground Officer → simplified field response workflow.
- Return to India map → navigate virtually to another sector/camera.

14. POSITIONING

Do not sell the idea as “AI detects people on CCTV.” Existing systems already do that. The stronger story is: **AI surveillance + India-wide geospatial command + role-specific operations + trusted evidence provenance/integrity + existing-CCTV compatibility + demo-ready low-connectivity architecture.**

BUILD PRINCIPLE	Make it look like a serious security product, make the demo reliable, and clearly separate prototype/demo capabilities from real-world deployment claims.